

Rank correlation and wavelength error in a fixed Rhomax model panel

Rewire Bio

Historical evidence report, 6 October 2026

Abstract

We document a fixed six-configuration panel for peak absorption wavelength on the FLIP2 Rhomax `by_wild_type` split. The preserved test set contains 184 rows, with 584 rows used for fitting and 116 held out for validation. Composition features with a fixed ridge head reach Spearman correlation about 0.418, but their mean absolute error is 24.36 nm versus 24.60 nm for a training-mean predictor: an improvement of only 0.24 nm. The reconstructed-group bootstrap resolves the direction of that difference while its magnitude remains below the historical 2 nm reporting threshold. Two frozen ESM-2 configurations have larger point errors; this is a comparison of fixed configurations, not a tuned comparison of model families. Candidate selection also depends on the objective: proximity to a desired wavelength band and maximising wavelength produce different shortlists. All numerical results in this report are imported from the preserved October 2026 study artifacts. The manuscript build checks their hashes and formats them; it does not constitute a new experiment or independent scientific reproduction.

1 Question and scope

The practical question is whether a model that ranks held-out sequences well also provides useful wavelength estimates or useful shortlists. These are separate outcomes. A rank statistic ignores absolute scale, while selecting near a target wavelength requires predictions whose values have operational meaning.

Rhomax is distributed through FLIP2, which provides explicit splits intended to assess protein prediction under distribution changes [FLIP2 authors, 2026]. The dataset README attributes the underlying measurements to Inoue et al. [2021]. We use the released assignment rather than constructing a new split. The endpoint is peak absorption wavelength in nanometres. No result establishes expression, photostability, ion transport, activation efficiency or cellular function. There is no new benchmark.

This report preserves the original fixed panel and exploratory analyses. It does not reinterpret upstream FLIP2 scores as comparable measurements: the archived source ledger records differences in target-scaling protocol. It also does not infer the performance of fine-tuned or larger models from the two frozen configurations tested.

2 Data and fixed methods

2.1 Data identity and separation

The pinned release is labelled `zenodo-18433203-v3`. Its decompressed CSV has SHA256 `8e78f6a1...a65f656f`; the complete digest and retrieval URLs are in the companion README and source ledger. Gzip framing differs between the mirror and Zenodo, so decompressed bytes define dataset identity. The CSV provides sequence, target, set and validation columns. It does not provide per-row wild-type identities, assay conditions or repeat-measurement identifiers. The upstream split name is therefore not sufficient to recover labelled biological groups from the CSV alone.

The archived audit reports no exact sequence overlap between the three splits. It also records duplicate sequences within training, including one pair differing by 19 nm in recorded target. This observation is not an estimate of general assay noise, and the archive does not establish whether the duplicates reflect repeat assays or repackaging. A supervised separation check also does not establish absence of overlap with the UniRef pretraining data used for the encoder [Meta FAIR, 2023]. Pretraining exposure remains undetermined.

2.2 Predictor configurations

The constant predictor is the mean of training targets. Composition-22 uses $\log(1 + L)$, 20 residue fractions and an unknown-residue fraction, which is zero on the pinned alphabet. Composition-40 retains the archived two-component construction: 20 residue fractions followed by 20 zero columns because Rhomax sequences contain no component delimiter. These are distinct historical configurations; the redundant columns are retained for fidelity.

The encoder configurations use ESM-2 8M (layer 6, 320 dimensions) and 35M (layer 12, 480 dimensions) [Lin et al., 2023, Meta FAIR, 2023]. Each sequence is represented by the frozen final-layer mean over residue positions, excluding start, end and padding tokens. Checkpoint bytes are pinned before loading. No encoder weights are trained in this panel.

All four feature representations feed a ridge head with $\alpha = 10$, an intercept, solver `auto`, tolerance 10^{-5} and maximum iteration count 10^6 . Features are not scaled. Targets are standardised using training rows only and mapped back to nanometres after prediction. Validation and test labels do not participate in head fitting or historical hyperparameter selection. Equal α does not impose equal effective regularisation across differently scaled representations.

The additional nearest-sequence control uses normalised Hamming distance among training sequences of equal length. Ties resolve to the lowest training CSV row. When there is no equal-length candidate, it predicts the training mean. This control uses no alignment and treats an insertion or deletion as a length mismatch; its limitations are intentional and explicit.

2.3 Metrics and exploratory uncertainty

We report mean absolute error (MAE), root mean squared error (RMSE), Spearman correlation and full-ranking NDCG. The NDCG relevance transform subtracts the minimum scored target, uses linear gains and averages predicted-score ties. Spearman is undefined for a constant predictor. Its NDCG is defined and can be large without sequence-specific skill.

The historical exploratory protocol reconstructs groups by single linkage among same-length

test sequences at at most 10% Hamming distance. It yields 37 groups, including 22 singletons; the largest has 30 rows. These are declared sequence clusters, not recovered wild-type labels. Paired bootstrap draws sample whole groups with replacement and score both predictors on the same sampled rows. The archived procedure uses 10,000 draws and seed 20261001. The point estimate is the full-test difference in MAE. A difference is called supported only when the percentile interval excludes zero and the absolute point difference reaches 2 nm. This is an exploratory reporting rule, not a multiplicity-adjusted confirmatory test, an equivalence test, or a universally justified biological tolerance.

3 Results

3.1 Rank advantage is a small error reduction

Table 1 records all configurations. Composition-22 reduces MAE by 0.24 nm relative to the constant. Its paired 95% bootstrap interval for candidate-minus-constant MAE is approximately $[-0.45, -0.05]$ nm, below zero but below the 2 nm reporting threshold in magnitude. Composition-40 behaves almost identically. A positive rank correlation alone therefore overstates the benefit for numerical wavelength estimation on this split.

Table 1: Historical held-out results on 184 test rows. Undefined correlation is not zero.

Configuration	MAE (nm)	RMSE (nm)	Spearman	NDCG
Training mean	24.60	32.50	undefined	0.9207
Composition-22	24.36	32.29	+0.418	0.9548
Composition-40	24.36	32.29	+0.418	0.9548
Nearest sequence	27.50	34.73	+0.338	0.9453
ESM-2 8M	28.90	41.23	-0.146	0.8965
ESM-2 35M	29.16	39.04	-0.222	0.9073

The two frozen ESM-2 configurations have larger MAE point estimates than the constant. Only the 35M configuration meets both parts of the historical difference rule; the 8M interval crosses zero. The nearest-sequence control also has an interval spanning zero. None of those intervals establishes equivalence. The panel does not select a tuned representation, tune ridge regularisation, or compare a common effective penalty.

The composition prediction spread explains the gap between ranking and numerical accuracy (Table 3). Composition-22 spans about 2.2 nm while the measured test standard deviation is about 32.15 nm. A narrow prediction distribution may order some rows while remaining close to a constant. The archived scale diagnostics also show that representation variances differ substantially under the same ridge penalty.

3.2 Distance strata are descriptive

The fixed distance bins are near ($d \leq 0.05$), mid ($0.05 < d \leq 0.25$), far ($0.25 < d < 1$), and no equal-length training sequence. There are 92 test rows in the final category. The near bin has only five rows; no row lies in the mid bin. The nearest-sequence control has the smallest near-bin MAE, while errors increase in more distant strata. Those within-bin comparisons have no paired interval and are descriptive, even when their row count meets the minimum of ten. The length gate

Rank correlation and wavelength error rank the same six configurations differently

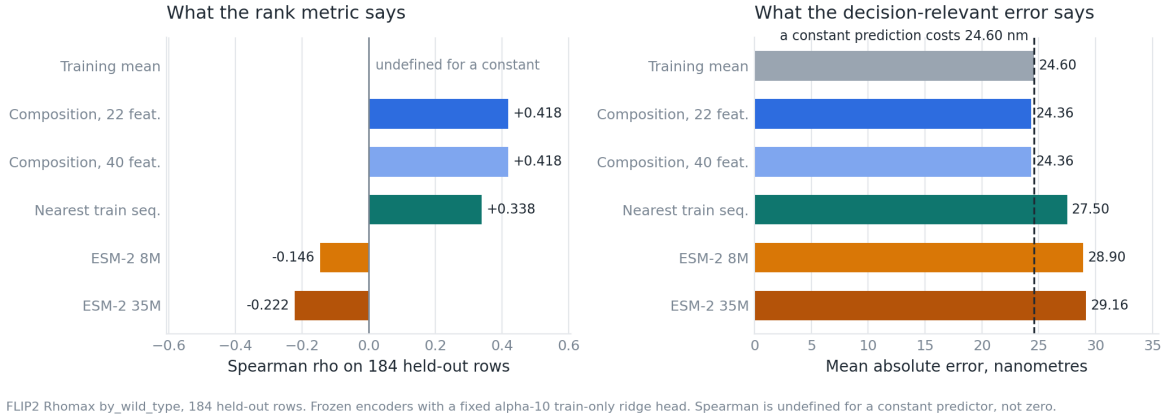


Figure 1: Original preserved rank-versus-error figure. Values describe the fixed historical panel.

can classify a close indel variant as lacking a neighbour, so the strata are not a general homology measure.

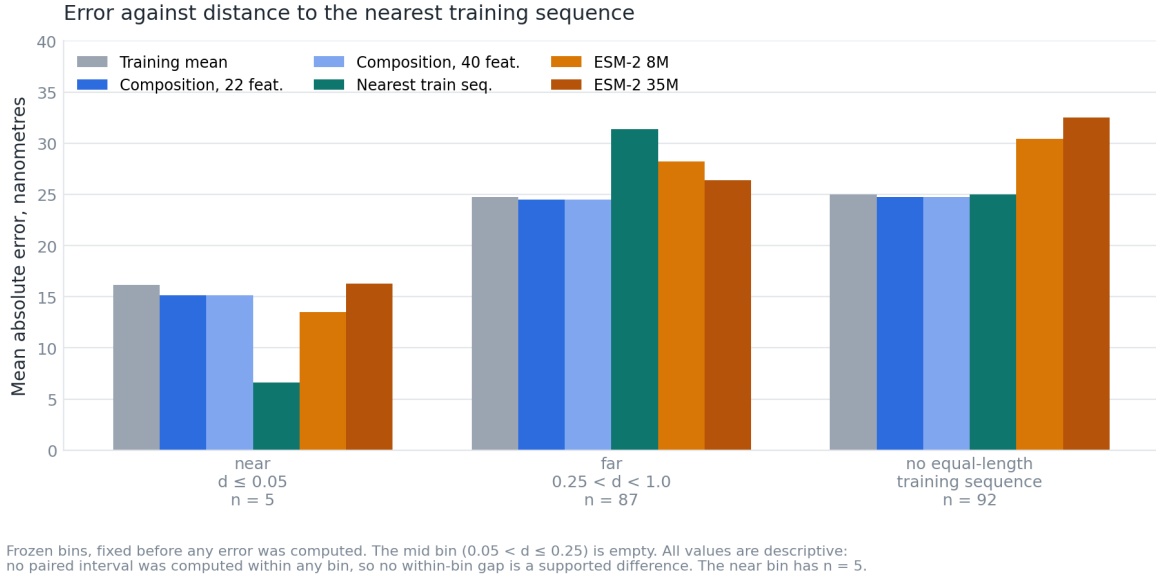


Figure 2: Original descriptive distance-stratum results. Absence of within-bin uncertainty limits comparative claims.

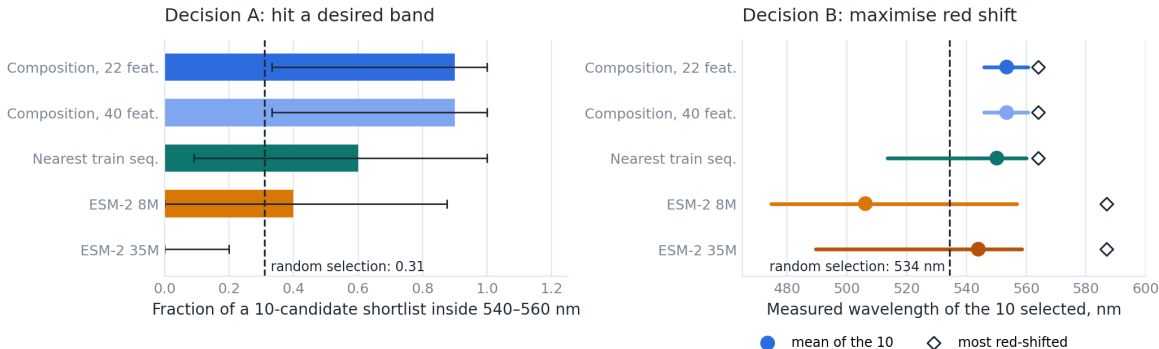
3.3 Two shortlist objectives

Decision A selects ten rows with predictions closest to 550 nm and measures the fraction whose observed wavelengths lie in $[540, 560]$ nm. Decision B selects the ten largest predictions and reports their mean observed wavelength. Ties use stable CSV order. A constant predictor has no defined shortlist under this policy.

Composition-22 puts 9 of 10 selected rows in the desired band, compared with an analytic uniform-random expectation of about 0.310. Its red-shift shortlist has mean measured wavelength 553.50 nm. These are different estimands; neither alone establishes accurate individual predictions. The post-hoc shortlist bootstrap resamples reconstructed groups and holds shortlist fraction fixed, so the realised depth varies from 3 to 21 for Composition-22. The resulting intervals are exploratory spreads, not repeated independent tests of a ten-sequence intervention.

A high-wavelength hit among ten candidates is also not sufficient evidence of skill. The archived finite-population calculation gives approximately 0.3283 probability that a uniform random ten-row shortlist includes at least one observation at or above 580 nm. This comparison conditions on the existing test population and does not predict prospective laboratory success.

Two different decisions, two different answers from the same six configurations



Capacity and band were fixed before evaluation. Intervals are post-hoc cluster bootstraps over a reconstructed grouping, not predeclared tests. A random top 10 contains a protein at or above 580 nm with probability 0.33, so ESM-2's 587 nm hit is within chance. The training mean has no shortlist at all: every one of the 184 rows ties.

Figure 3: Original shortlist figure. Desired-band precision and mean selected wavelength answer different questions.

3.4 Post-hoc interval feasibility

The preserved probe sets a symmetric half-width to the empirical 90th percentile of absolute validation residuals, using the higher order-statistic interpolation rule. It then reports test coverage. The procedure is descriptive; it is not asserted to be an exact finite-sample conformal interval. Observed coverage is below the nominal 90% for every configuration (Table 7). Row-level Wilson intervals are retained for fidelity, but their independent-row assumption does not account for reconstructed group dependence. They must not be interpreted as background-level coverage guarantees. The validation set is a single held-out collection, so coverage under new background shifts remains unestablished.

4 Limitations and interpretation

The main conclusion is conditional: on this split and these fixed configurations, the composition ranking advantage buys little absolute-error improvement over a constant. The evidence does

not imply that protein language models are generally inferior, that ridge is optimally regularised here, or that the preferred shortlist transfers to a new assay. The bounded panel lacks alignment-based controls, tuned or fine-tuned models, repeated labelled background splits and a prospective experimental validation.

Missing wild-type and assay metadata restrict causal interpretation of error differences. Single-linkage reconstructed groups can join chains of near neighbours and are an imperfect dependence model. No adjusted family-wise significance claim is made across the multiple model comparisons. The observed results cannot separate measurement variation, representation limitations and distribution shift into independent effects.

A future study could predeclare model tuning, an alignment policy, independent background evaluation and a decision-specific utility measure. Those are proposals; no such experiment is reported here.

5 Provenance, maintenance and availability

The repository preserves the original detailed article, published text, figures, code archive and result archive. Two original prediction NPZ files omitted from the published tarball were subsequently recovered unchanged; their recovery manifest records hashes and byte identity with the retained run directory. The publication archives themselves remain unchanged.

The archived receipt records a historical reimplementing matching the earlier five configuration rank metrics. This manuscript reports that receipt as historical evidence; it does not convert it into a new independent reproduction of the current repository. The current paper build verifies archive and member digests, checks printed article values, generates tables and compiles this document. The generated claim ledger maps findings to archived JSON fields. No training or new scientific analysis is required for the build. The current root research harness remains unapproved and disabled.

The October 2026 maintenance audit repaired installation portability, test coverage, invalid numeric arguments, prediction metadata checks and embedding cache provenance. Those guards do not change the historical settings or measurements. Existing caches without provenance require an explicit refresh before future use. Original downloaded code remains an archival snapshot; current execution instructions live in the maintained companion README.

All source measurements retain upstream licensing and attribution. Checkpoints are retrieved from upstream sources and are not redistributed. Repository materials are at <https://github.com/rewire-bio/rhodopsin-model-selection>.

References

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- K. Inoue, M. Karasuyama, R. Nakamura, et al. Exploration of natural red-shifted rhodopsins using a machine learning-based bayesian experimental design. *Communications Biology*, 4:362, 2021. doi: 10.1038/s42003-021-01878-9. URL <https://doi.org/10.1038/s42003-021-01878-9>. Measurement attribution follows the FLIP2 Rhomax dataset README.
- Zeming Lin, Halil Akin, Roshan Rao, Brian Hie, et al. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379:1123–1130, 2023. doi: 10.1126/science.ade2574. URL <https://doi.org/10.1126/science.ade2574>. Bibliographic identity verified against the official ESM repository README.
- Meta FAIR. Evolutionary scale modeling: Esm repository, 2023. URL <https://github.com/facebookresearch/esm/tree/2b369911bb5b4b0dda914521b9475cad1656b2ac>. Revision 2b369911bb5b4b0dda914521b9475cad1656b2ac; retrieved 6 October 2026.

A Complete numerical supplement

The following tables are generated directly from the preserved result archive. No original result artifact is dropped: the mapping in `evidence/paper-migration-map.json` identifies original tables, figures and supplemental files. The detailed original article remains available for its full historical narrative and bibliography.

Table 2: Full panel and historical paired difference interpretation. MAE differences are candidate minus training mean.

Configuration	Spearman	NDCG	MAE	RMSE	Δ MAE	95% interval	Interpretation
Composition, 22 features	+0.418	0.9548	24.36	32.29	−0.24	[−0.45, −0.05]	direction resolved, below the 2 nm threshold
Composition, 40 features	+0.418	0.9548	24.36	32.29	−0.24	[−0.45, −0.05]	direction resolved, below the 2 nm threshold
Training mean	undefined	0.9207	24.60	32.50	baseline	n/a	comparator
Nearest training sequence	+0.338	0.9453	27.50	34.73	+2.90	[−7.86, +12.21]	no supported difference
ESM-2 8M, layer 6, residue mean	−0.146	0.8965	28.90	41.23	+4.30	[−0.58, +11.26]	no supported difference
ESM-2 35M, layer 12, residue mean	−0.222	0.9073	29.16	39.04	+4.56	[+0.80, +9.25]	supported: worse than a constant

Table 3: Prediction spread; the measured test standard deviation is 32.15 nm.

Configuration	SD (nm)	Span (nm)	SD/measured	Pearson
Training mean	0.00	0.0	0%	undefined
Composition-22	0.65	2.2	2.0%	+0.3621
Composition-40	0.65	2.2	2.0%	+0.3621
ESM-2 35M	13.66	57.2	42.5%	-0.3488
ESM-2 8M	15.73	54.7	48.9%	-0.3120
Nearest sequence	19.69	75.0	61.2%	+0.2550

Table 4: Representation scales under the same $\alpha = 10$. This is not equal effective regularisation.

Representation	Dimension	Zero columns	Total variance	Variance/ α
Composition-22	22	1	0.010571	0.00105706
Composition-40	40	20	0.003450	0.00034495
ESM-2 35M	480	0	0.817550	0.08175505
ESM-2 8M	320	0	0.745695	0.07456947

Table 5: Desired-band shortlist. Spread intervals are post hoc; random selection is an analytic expectation.

Configuration	In band	Precision	Deviation (nm)	95% spread
Composition-22	9/10	0.90	6.30	[0.33, 1.00]
Composition-40	9/10	0.90	6.30	[0.33, 1.00]
Nearest sequence	6/10	0.60	17.40	[0.09, 1.00]
ESM-2 8M	4/10	0.40	23.60	[0.00, 0.88]
ESM-2 35M	0/10	0.00	69.80	[0.00, 0.20]
Training mean	no shortlist	n/a	n/a	n/a
Uniform random (analytic)	n/a	0.3098	26.31	n/a

Table 6: Red-shift shortlist. The oracle is a retrospective upper bound, not a deployable predictor.

Configuration	Mean (nm)	Maximum (nm)	95% spread (nm)
Oracle (best possible top 10)	583.30	589	n/a
Composition-22	553.50	564	[546.0, 560.6]
Composition-40	553.50	564	[546.0, 560.6]
Nearest sequence	550.20	564	[513.6, 560.0]
ESM-2 35M	544.00	587	[489.5, 558.6]
Uniform random (analytic)	534.43	n/a	n/a
ESM-2 8M	506.20	587	[474.7, 556.8]
Training mean	no shortlist	n/a	n/a

Table 7: Archived validation-residual interval feasibility. Wilson intervals treat rows as independent and are descriptive only.

Configuration	Half-width (nm)	Coverage	Wilson 95%	Nominal
Training mean	37.7	0.772	[0.706, 0.826]	0.90
Composition-22	38.1	0.783	[0.718, 0.836]	0.90
Composition-40	38.1	0.783	[0.718, 0.836]	0.90
Nearest sequence	41.7	0.788	[0.723, 0.841]	0.90
ESM-2 35M	43.4	0.766	[0.700, 0.822]	0.90
ESM-2 8M	47.5	0.837	[0.777, 0.883]	0.90

Table 8: Distance bins and descriptive MAE (nm). Composition-40 agrees with Composition-22 at the shown precision.

Bin	n	$n \geq 10$	Target range	Nearest	ESM-8M	Comp.-22	Constant	ESM-35M
near, $d \leq 0.05$	5	no	523–560	6.60	13.49	15.11	16.16	16.29
mid, $0.05 < d \leq 0.25$	0	no, empty	n/a	n/a	n/a	n/a	n/a	n/a
far, $0.25 < d < 1.0$	87	yes	470–589	31.38	28.18	24.47	24.71	26.38
no equal-length training sequence	92	yes	436–587	24.97	30.41	24.76	24.97	32.49

Table 9: Example test rows from the original article. Digests are local sequence identifiers, not accessions. All wavelengths are in nm.

Sequence digest	CSV row	Length	Measured	Distance	Neighbour digest	Nearest	Comp.-22	ESM-35M
1126648311d3	187	280	523	0.0179	37b98f5e1e0a	525.00	538.81	527.60
78c26b0b585d	223	262	558	0.0038	18414205e448	551.00	540.82	575.86
1937f73b396e	224	262	557	0.0076	18414205e448	551.00	540.83	574.32
c28e8cbcbc78	225	262	560	0.0038	72fa783e9afa	575.00	540.78	575.38
848b6cf05e5c	226	262	548	0.0076	18414205e448	551.00	540.82	574.27
781ca2b1c04b	166	350	587	1.0000, no match	none	539.73	539.56	562.94

Table 10: Paired reconstructed-group bootstrap details. The reporting threshold is 2 nm in absolute point difference.

Configuration	ΔMAE	95% interval	Bias	Fraction better	Excludes 0	Meets 2 nm	Supported
Composition-22	-0.2438	$[-0.4530, -0.0507]$	-0.0070	0.9940	yes	no	no
Composition-40	-0.2436	$[-0.4530, -0.0503]$	-0.0069	0.9941	yes	no	no
Nearest sequence	+2.8957	$[-7.8552, +12.2116]$	-0.1261	0.2738	no	yes	no
ESM-2 8M	+4.2956	$[-0.5835, +11.2554]$	+0.2050	0.0499	no	yes	no
ESM-2 35M	+4.5596	$[+0.8044, +9.2482]$	+0.1873	0.0075	yes	yes	yes

Table 11: Per-reconstructed-group MAE (nm), descriptive only.

Group	n	Constant	Comp.-22	Nearest	ESM-8M	ESM-35M
bg01	30	15.94	15.75	27.80	12.50	11.90
bg02	27	11.94	12.24	11.94	19.86	19.15
bg03	23	21.65	21.14	21.65	15.65	19.57
bg04	19	10.33	10.20	41.26	9.30	13.01
bg05	14	62.80	62.63	23.07	93.53	75.26
bg06	12	32.77	32.05	32.77	37.71	51.30
bg07	8	28.98	29.31	11.75	33.81	29.25
bg08	5	66.73	66.57	66.73	92.34	87.80
bg09	5	44.07	43.64	83.80	43.27	54.71
bg10	4	5.37	5.36	5.37	30.88	25.40
bg11	4	16.02	14.94	7.75	14.95	19.21
bg12	3	19.93	18.66	11.67	12.90	21.80
bg13	3	6.27	5.00	46.00	6.10	7.25
bg14	3	46.40	45.94	46.40	61.94	61.35
bg15	2	10.73	10.05	10.73	5.07	14.17
bg16	1	47.27	47.44	47.27	8.02	24.06
bg17	1	44.73	44.46	44.73	68.73	69.18
bg18	1	86.73	86.32	86.73	109.30	106.61
bg19	1	8.73	8.82	8.73	41.12	40.23
bg20	1	16.73	15.81	2.00	7.65	4.60
bg21	1	4.73	5.09	4.73	5.06	3.95
bg22	1	12.27	11.39	12.27	22.36	25.84
bg23	1	12.27	11.63	12.27	8.37	9.23
bg24	1	38.27	37.66	38.27	39.89	52.07
bg25	1	33.73	33.51	33.73	30.93	19.96
bg26	1	15.27	14.05	55.00	9.86	20.27
bg27	1	19.73	19.06	11.00	8.17	8.38
bg28	1	2.27	1.85	2.27	10.76	16.36
bg29	1	54.73	55.26	54.73	46.50	36.63
bg30	1	14.73	14.18	14.73	1.10	1.23
bg31	1	5.73	6.52	5.73	15.53	14.04
bg32	1	5.27	4.85	2.00	19.74	13.64
bg33	1	1.73	1.40	1.73	1.35	12.79
bg34	1	95.73	95.49	95.73	124.07	107.77
bg35	1	103.73	103.53	103.73	135.15	113.80
bg36	1	12.73	12.62	21.00	4.35	0.03
bg37	1	11.73	11.22	14.00	4.53	0.13

Table 12: Historical reproduction receipt: archived rank metrics and differences reported by the original companion run. Not a new reproduction.

Configuration	Spearman	Difference	NDCG	Difference	Status
Training mean	undefined	n/a	0.92066675	0	match
Composition-40	+0.41818225	0	0.95481994	0	match
Composition-22	+0.41798958	0	0.95481549	0	match
ESM-2 8M	-0.14635068	0	0.89647998	0	match
ESM-2 35M	-0.22175950	0	0.90725802	0	match

Table 13: Historical single-run feature/embedding and fit/prediction times (seconds). The receipt reports 49.2s total and 1.42 GiB peak RSS.

Configuration	Features/embedding	Fit/predict
Training mean	–	0.0001
Composition-22	0.0038	0.0010
Composition-40	0.0053	0.0014
Nearest sequence	–	0.0152
ESM-2 8M	13.2	0.0441
ESM-2 35M	35.2	0.0074

Table 14: Finite-population random shortlist references, capacity ten of 184.

Threshold	Population hits	Probability of at least one
≥ 570 nm	26	0.7910
≥ 580 nm	7	0.3283
≥ 585 nm	4	0.2018

Table 15: Measured wavelengths in the original red-shift shortlists, in predicted rank order.

Configuration	1	2	3	4	5	6	7	8	9	10
Composition-22	548	558	541	557	549	564	555	557	558	548
Nearest sequence	560	558	557	548	558	564	557	527	545	528
ESM-2 8M	587	548	531	470	436	490	480	480	560	480
ESM-2 35M	558	560	557	548	531	495	530	539	587	535

B Additional preserved diagnostics

The NDCG finite-population reference gives reverse-order score 0.8131, random permutation mean 0.9207 and percentile range [0.8975, 0.9403] across 10,000 draws (seed 20261002). These are preserved diagnostics, not new permutation tests. The exact constant-score NDCG is determined by tie averaging and the observed label distribution.

The archived environment records Python 3.11.13, NumPy 1.26.4, SciPy 1.15.3, scikit-learn 1.8.0 and torch 2.2.2. A historical CPU/MPS embedding comparison on 40 sequences recorded maximum absolute difference 1.4e-06; that small device check is not a full cross-platform reproduction.

The best-ranking configuration barely moves; the worst moves further, and the wrong way

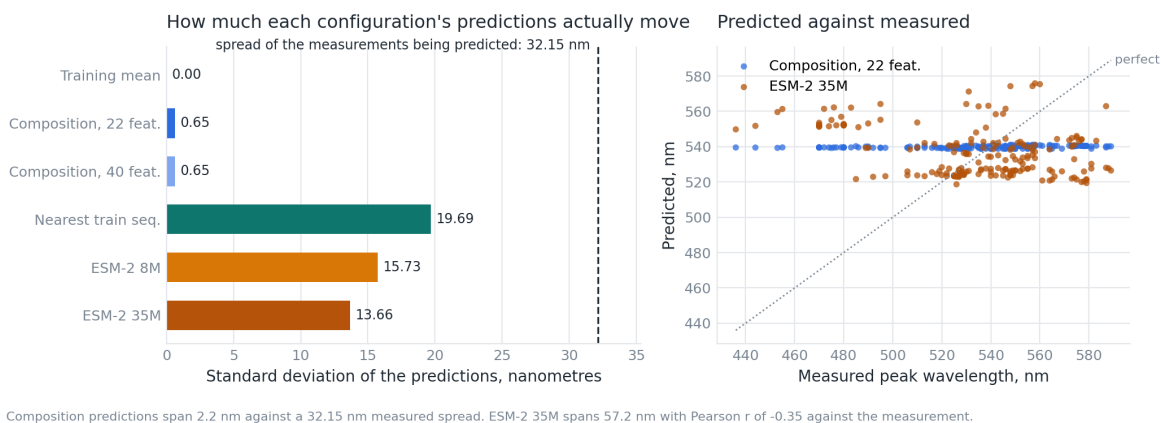


Figure 4: Original preserved prediction-spread figure. The source SVG decision diagram and its D2 source remain in the archive mapping.